

# **EduGenie – Module Design**

**Project: EduGenie – Google Gemini Powered Learning Assistant**

## **1. Introduction**

EduGenie is divided into separate modules. Each module performs a specific learning function, making the application easier to develop, maintain, and extend.

## **2. Module Structure**

### **2.1 Q&A Module**

**Purpose:**

Provides answers to academic questions entered by the student.

**Input:**

Student's question.

**Processing:**

The question is sent for AI-based processing.

**Output:**

A clear and concise answer.

### **2.2 Explanation Module**

**Purpose:**

Explains difficult educational concepts in simple language.

**Input:**

Topic or concept provided by the student.

**Processing:**

The concept is processed to generate an easy-to-understand explanation.

**Output:**

A simplified explanation.

### **2.3 Quiz Module**

**Purpose:**

Generates quiz questions for student practice.

**Input:**

Topic or subject provided by the student.

**Processing:**

The system generates questions based on the selected topic.

**Output:**

An educational quiz.

## **2.4 Summary Module**

**Purpose:**

Summarizes educational passages and text.

**Input:**

Educational text provided by the student.

**Processing:**

Important information is identified and condensed.

**Output:**

A concise summary.

## **2.5 Learning Path Module**

**Purpose:**

Provides learning recommendations for a selected topic.

**Input:**

Topic or learning requirement.

**Processing:**

The system generates suitable learning recommendations.

**Output:**

A recommended learning path.

## **3. Backend Module**

The FastAPI backend manages communication between the frontend, learning modules, and AI service. It receives requests, processes them, and returns results to the user interface.

## **4. Frontend Module**

The frontend provides the interface through which students can select a learning task, enter information, submit requests, and view results.

## **5. Conclusion**

The modular design of EduGenie separates different learning functions into manageable components. This makes the application easier to maintain and allows new features to be added in the future.